THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE UNIT TEST SET B

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The atomic theory explains: 1

(a) what 1 kg is **(b)** how molecules are formed
(c) why clouds look dark **(d)** how to measure time

2. The main purpose of making a rough estimate is to: 1

(a) replace all exact measurements **(b)** make science harder
(c) check whether an answer is reasonable **(d)** avoid using units

3. 'Scientific theory' differs from the everyday word 'theory' because the scientific one: 1

(a) is backed by careful testing and evidence
(b) is only an idea in someone's mind
(c) has never been tested
(d) is the same as a belief

4. Which technique is used to separate the different dyes present in black ink? 1

(a) paper chromatography **(b)** sublimation
(c) distillation **(d)** decantation

5. A solution that cannot dissolve any more solute at a given temperature is called: 1

(a) a suspension **(b)** a saturated solution
(c) a colloid **(d)** a dilute solution

6. Which of the following sublimes on heating? 1

(a) iron filings **(b)** common salt **(c)** naphthalene **(d)** sand

7. The chemical bond formed by sharing of electrons is called: 1

(a) covalent bond **(b)** metallic bond only
(c) ionic bond **(d)** hydrogen bond

8. In an oxygen molecule O_2 the two atoms are joined by: 1

(a) a triple bond **(b)** an ionic bond **(c)** a double bond **(d)** a single bond

9. The chemical formula of calcium chloride is: 1

(a) Ca_2Cl **(b)** $CaCl_2$ **(c)** Ca_2Cl_2 **(d)** $CaCl$

10. The nucleus is about _____ times smaller than the atom. 1

(a) 10 **(b)** 10^2
(c) 10^{10} **(d)** 10^5 (one lakh)

11. An atom of chlorine has atomic number 17 and mass number 35. The number of neutrons in it is: 1

(a) 19 **(b)** 35 **(c)** 18 **(d)** 17

12. The neutron was discovered by: 1

(a) James Chadwick **(b)** Niels Bohr **(c)** J. J. Thomson **(d)** Goldstein

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SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. State one consequence of mixing up pounds and kilograms as described in the aircraft incident. 2

14. What is a saturated solution? 2

15. If 12 g carbon combines with 32 g oxygen, how much CO_2 forms from 6 g carbon? 2

16. State the Law of Conservation of Mass. 2

17. What did most α-particles do in the gold foil experiment and what does it show? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. A student wishes to model the time taken to cycle from school to home. Which details would she keep and which ignore? Why? 3

19. Calculate how much zinc oxide is present in 300 g of a talcum powder containing 4 % m/m zinc oxide. 3

20. How do Dalton's postulates explain 3
(i) the law of conservation of mass
(ii) the law of constant proportions?

21. If the diameter of an atom is 10^{-10} m and of its nucleus 10^{-15} m, how many times smaller is the nucleus? Use the cricket ground analogy. 3

22. Explain how Meghna could test Varsha's prediction of rain with measurable evidence. Give three questions. 3

OR

Explain with an example how models are built and why assumptions are made. 3

23. A student mixes acetone (b.p. 56 °C) and water (b.p. 100 °C). Explain how to separate them and why evaporation cannot be used. 3

OR

Vinegar contains 5 % v/v acetic acid. How much glacial acetic acid is needed to make 200 mL of vinegar? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. **Case study:** A student wants to analyse a pressure cooker using ideas from different branches of science. 5
(i) Which physics idea is involved?
(ii) Which chemical or biological idea?
(iii) How does mathematics help?
(iv) Why can the cooker not be studied by one branch alone?

OR

Case study: A passenger aircraft ran out of fuel because the ground crew used the fuel density in pounds per litre instead of kilograms per litre. 5
(i) What type of error is this?
(ii) Why did the plane have to glide?
(iii) How do SI units help?
(iv) Name two quantities whose units must always be checked in such calculations.

25. **Case study:** Sodium chloride contains sodium and chlorine in the ratio 23 : 35.5 by mass. 5
(i) How much chlorine reacts with 46 g of sodium?
(ii) How much NaCl forms?
(iii) How much sodium is in 117 g of NaCl?
(iv) Which law helps?

OR

(i) State the postulates of Dalton's atomic theory. 5
(ii) How does it explain the laws of chemical combination?
(iii) Which postulate was later shown to be incorrect and why?

END OF QUESTION PAPER — **Total: 50 Marks**